

European Journal of Engineering Education

ISSN: 0304-3797 (Print) 1469-5898 (Online) Journal homepage: www.tandfonline.com/journals/ceee20

Understanding the experiences of academic staff in fostering engineering students' digital capabilities: a case study using Activity Theory

Tamsyn M. Smith

To cite this article: Tamsyn M. Smith (09 Apr 2026): Understanding the experiences of academic staff in fostering engineering students' digital capabilities: a case study using Activity Theory, European Journal of Engineering Education, DOI: 10.1080/03043797.2026.2651418

To link to this article: https://doi.org/10.1080/03043797.2026.2651418

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 09 Apr 2026.

Submit your article to this journal

Article views: 518

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ceee20

EUROPEAN JOURNAL OF ENGINEERING EDUCATION
https://doi.org/10.1080/03043797.2026.2651418

Taylor & Francis
by informa

OPEN ACCESS Check for updates

Understanding the experiences of academic staff in fostering engineering students' digital capabilities: a case study using Activity Theory

Tamsyn M. Smith

Digital Learning, iSolutions, University of Southampton, Southampton, UK; Department of Educational Research, Lancaster University, Lancaster, UK

ABSTRACT

This case study investigates academics' experiences when developing engineering students' digital capabilities within the context of workload intensification at a UK research-intensive university. Using Activity Theory as a theoretical framework, semi-structured interviews with six academics explored their teaching practices and the systemic tensions they encounter. Analysis identified secondary contradictions between available *Tools* (such as rubrics and' hardware) and the *Object* of producing digitally capable graduates, exacerbated by time constraints from the *Division of Labour*. Quaternary contradictions emerged between academics' and management's activity systems regarding guidance provision, resource allocation, and balancing personalisation against massification pressures. Key findings reveal that academics recognise digital capabilities' importance for employability but struggle with implicit skill expectations, inadequate assessment criteria for technical competencies, and insufficient time for curriculum redesign. Unexpectedly, academics sought clearer guidance rather than autonomy. This research illuminates barriers to integrating discipline-specific digital competencies – including computer-aided design proficiency and computational methods – into engineering curricula.

ARTICLE HISTORY
Received 22 May 2025
Accepted 22 March 2026

KEYWORDS
Digital capabilities; digital literacy; engineering; workload intensification; Activity Theory

Introduction

The development of students' digital capabilities has become increasingly prioritised in higher education (HE) globally. Digital capabilities – defined as the skills, knowledge, and attitudes that enable individuals to thrive in an increasingly digital world (Jisc 2019) – are now considered essential components of a comprehensive educational experience. This emphasis on digital capabilities reflects broader societal shifts towards digitalisation, including the growing influence of generative artificial intelligence (GenAI) and emerging technologies on professional practice.

Engineering education faces particular challenges in integrating digital capabilities. Professional accreditation bodies worldwide have embedded digital competency requirements within their frameworks, creating mandatory expectations that transcend institutional choice. At the international level, the International Engineering Alliance widened its Graduate Attributes profile in 2020 'to emphasise digital literacy, [and] data analysis' (International Engineering Alliance 2021). The Washington Accord mandates that graduates 'create, select and apply appropriate techniques,

CONTACT Tamsyn M. Smith t.m.smith@soton.ac.uk

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

resources, and modern engineering and IT tools […] to complex engineering problems' (International Engineering Alliance 2025), with similar requirements for Sydney and Dublin Accord graduates (International Engineering Alliance 2021). These international standards create baseline expectations that national and regional frameworks must meet or exceed.

European frameworks reinforce these expectations. The European Network for Accreditation of Engineering Education (ENAEE) established EUR-ACE® standards requiring graduates to 'conduct searches of literature, to consult and critically use databases and other sources of information, to carry out simulation' and to possess 'knowledge and understanding of mathematics, computing and other basic sciences' (European Network for Accreditation of Engineering Education 2024). National accreditation bodies such as Germany's ASIIN (New Jersey Institute of Technology 2025) and Switzerland's AAQ (2025) can award the EUR-ACE® label, demonstrating pan-European commitment to consistent digital competency standards. In the UK, the Engineering Council's Accreditation of Higher Education Programmes (AHEP4) expects engineering graduates to be able to use, understand, and critique digital tools and computational methods (Engineering Council 2020). These frameworks collectively position digital capabilities as fundamental to contemporary engineering identity and professional practice. Unlike generic digital literacy, they specify technical competencies – including computer–aided design (CAD), simulation software, data analysis tools, and computational methods – that are essential to professional engineering practice rather than transferable skills across disciplines. However, the practical implementation of digital capabilities development within disciplinary teaching remains challenging, and despite the clarity of these frameworks, embedding such specialised tools and methods consistently within engineering curricula continues to present significant difficulties.

Implementing these digital capability requirements occurs within a context of significant structural pressures affecting academics across the sector. The massification of HE – characterised by rapidly expanding student numbers without proportionate increases in resources – has intensified academic workloads internationally (Hartman and Darab 2012; Tessema 2009). This expansion has occurred alongside the introduction of market-driven approaches that emphasise measured outputs, accountability, and institutional ranking (Larsen and Brandenburg 2023; Tham and Holland 2018), contributing to decreased faculty autonomy and an increased perception of managerial control (Blackmore 2014; Dowling-Hetherington 2014).

Academics across the sector navigate employment conditions marked by precarity and intensification. Precarious academic employment – encompassing temporary, part-time, hourly-paid, and zero-hours contracts – has grown exponentially in response to structural and fiscal changes within universities (Bryson 2004). This phenomenon is widespread internationally: in the UK, 54% of all academic staff and 49% of teaching staff are employed on insecure contracts (University and College Union 2016), whilst across Europe, temporary employment rates in academia substantially exceed national averages, with academics in several countries explicitly excluded from labour protections limiting contract duration (Dixon 2024; Molnárfi 2022; Organisation for Economic Co-operation and Development 2024). Meanwhile, teaching-only staff numbers at UK universities increased by more than 80% between 2005–2006 and 2018-2019, compared with only 16% growth in traditional teaching and research roles (Wolf and Jenkins 2021).

Academics must now incorporate multiple competing initiatives – including sustainability, inclusivity, employability, and digital capabilities – into already crowded curricula, whilst managing increased administrative burden (Blackmore 2014; Griffin 2022; Papadopoulos 2017) and the well-documented impacts on work-life balance and wellbeing (Griffin 2022; Houston, Meyer, and Paewai 2006; Kenny 2016; Kinman and Jones 2003; Vostal 2015). With graduate outcomes under increasing scrutiny (Lewis 2023; Van Essen-Fishman, Morgan, and Madrigal 2023), academics face pressure to demonstrate measurable outcomes whilst managing escalating demands. Those carrying the heaviest teaching responsibilities are particularly exposed to these pressures.

This research investigates the systemic tensions and contradictions that arise when academics attempt to implement digital capabilities development within engineering education. This study employs Activity Theory as its theoretical framework, which provides a structured approach to

analysing complex systems of human activity and their inherent contradictions (Engeström, 2014). Activity Theory posits that human actions are best understood within their social and cultural contexts, examining the interactions between *Subjects* (individuals), *Tools* (mediating artefacts), *Rules*, *Community*, *Division of Labour*, and *Objects*.

Literature review

This narrative review examines scholarship on digital capabilities integration in HE, drawing on engineering education research, digital capabilities frameworks, and higher education studies to contextualise the research focus.

Digital capability expectations in engineering education

Digital capabilities in HE encompass the skills to learn, work and flourish with digital technology in contemporary society (Saienko, Kurysh, and Siliutina 2022; Traxler 2018; Varga-Atkins 2020). The transfer of responsibility for employability to HE institutions, alongside increased supply of graduates, underscores the importance of integrating digital capabilities into curricula (Austen et al. 2016; Bennett et al. 2022; Cheng et al. 2021).

Engineering education faces specific expectations regarding digital capabilities. Scholars emphasise the urgency of preparing students for changing labour markets in the context of the Fourth Industrial Revolution (Aldhaen 2023; Blau, Shamir-Inbal, and Avdiel 2020; Cheng et al. 2021; Jones et al. 2020; Jugembayeva and Murzagaliyeva 2022; Morgan, Sibson, and Jackson 2022; Scott 2015a). This includes developing digital literacy, critical thinking, problem-solving skills, and adaptability to new technologies including AI and machine learning applications (Klochkova et al. 2020).

The emergence of GenAI has added new dimensions to digital capability requirements, forcing engineering educators to reconsider assessment strategies and the nature of digital skills development (Goswami and Souppez 2024; Qadir 2023; World Economic Forum 2024). These tools offer potential for enhanced learning experiences but expose significant gaps in students' critical evaluation skills and ability to verify information accuracy, emphasising that digital capabilities extend beyond operational competence to encompass sophisticated cognitive abilities, including assessment and appropriate application of AI-generated content (Goswami and Souppez 2024; Qadir 2023). However, despite recognition of the importance of these capabilities, the literature reveals limited evidence of systematic integration within disciplinary contexts.

Challenges for educators in integrating digital capabilities in curricula

Scholarship in this area suggests a range of challenges for educators seeking to integrate digital capabilities into curricula.

Limited practical guidance for implementation and alignment with educational frameworks

Whilst the literature emphasises the significance of digital competencies for students, it provides limited guidance for educators on how these capabilities can be developed in practice (Saienko, Kurysh, and Siliutina 2022; Scott 2015a; Wells 2023). This gap has become more pronounced with the rapid proliferation of AI tools, where significant disparities exist between students' assumed digital proficiency and their actual academic and professional capabilities, particularly in areas requiring information evaluation and critical analysis (Morgan, Sibson, and Jackson 2022). A related challenge concerns the alignment of digital literacy initiatives with broader educational objectives, which requires integration with frameworks such as Education 4.0 (World Economic Forum 2024), promoting data skills, critical thinking, and problem-solving alongside digital capabilities, so that students can tackle real-world challenges (Atenas, Havemann, and Timmermann 2020; Blau, Shamir-Inbal, and Avdiel 2020; Ramírez-Montoya et al. 2021; Saienko, Kurysh, and Siliutina 2022;

Scott 2015b; Sissons and Mulrennan 2022). However, the literature lacks examples of successful alignment within engineering curricula, and the rapidly evolving technological landscape underscores the need for adaptable frameworks that can accommodate emerging technologies whilst maintaining focus on fundamental cognitive and evaluative competencies (Goswami and Souppez 2024).

Professional development requirements and institutional barriers

Employability and workforce readiness demands require continuous professional development for educators to develop digital competence and integrate digital capabilities effectively (Masenya 2021; Ostanina et al. 2023; Saffron Powell and Varga-Atkins 2013). Evidence from institutional provision suggests that professional development opportunities are indeed available; however, academics may lack sufficient time to engage with these resources. The emergence of GenAI technologies has further intensified this challenge, requiring educators to develop competencies in teaching students to evaluate AI-generated content critically and use such tools responsibly (Qadir 2023). These professional development demands are compounded by wider institutional and pedagogical barriers that educators face when attempting effective integration, including a lack of confidence and time, insufficient institutional training, support and resources, resistance to change, a need for targeted interventions, assumptions about students' digital capabilities, and pedagogical concerns relating to disciplinary context (Cheng et al. 2021; Jones et al. 2020; Masenya 2021; Morgan, Sibson, and Jackson 2022; Murray, Pérez, and Fluker 2022). These barriers become more pronounced in the context of AI-generated content, where students may lack the foundational critical thinking skills necessary to identify misinformation or assess the quality and bias of algorithmically produced information (Qadir 2023).

Tensions between digital capabilities and disciplinary content coverage

Finally, the literature reveals tensions experienced by educators between the recommended practice of embedding digital capabilities within curricula and maintaining disciplinary content coverage (Armah and Westhuizen 2020; Varga-Atkins 2020). Student digital capabilities compete with other priorities when academics design teaching, yet contradictions in practice remain poorly understood. These tensions are likely to intensify as emerging technologies such as GenAI necessitate explicit teaching of new literacies within already crowded curricula (Goswami and Souppez 2024).

Summary and research focus

The literature on digital capabilities integration shows significant challenges facing academics, but a critical gap remains in understanding how these pressures specifically affect academics' ability to integrate digital capabilities within discipline-specific teaching activities. This gap is acute in engineering, where technical digital competencies – including CAD proficiency, computational methods, and simulation techniques – are intrinsic to professional practice rather than supplementary skills, and where large cohort sizes typical of engineering programmes intensify delivery challenges. Furthermore, whilst the literature documents various barriers to digital capabilities integration, the systemic tensions and contradictions that underpin these challenges remain under-theorised. Activity Theory provides an analytical framework for examining these multi-level tensions systematically.

This study therefore investigates how academics engage with developing students' digital capabilities within their teaching practice, and what systemic tensions emerge in this process. The research addresses two questions.

Research questions

The research questions, informed by the gaps identified in the literature, examine academics' engagement with digital capabilities in their teaching practice:

RQ 1. How do academics engage with student digital capabilities in their teaching?

RQ 2. What tensions and contradictions do academics experience when attempting to integrate digital capabilities within their teaching activities?

Theoretical framework

Activity Theory provides a comprehensive framework for examining how human activities unfold within their social and cultural contexts. Activity Theory understands activities as purposeful, collective practices that are mediated by artefacts and shaped by social and institutional structures. A central tenet is that activity is motivated by an objective – a goal or purpose that drives the system forward. Activity Theory illustrates how human behaviours are situated within institutions, with activity systems embodying multiple viewpoints and evolving over time (Otoo 2020). This perspective makes Activity Theory suited to examining complex educational practices where multiple stakeholders, technologies, and institutional factors interact.

Overview of activity theory components

Activity Theory has its roots in Vygotsky's 1920s sociocultural work on mediated human development, which proposed that *Tools* mediate between an individual and their goal. Leontiev subsequently extended this to encompass collective activity systems, and Engeström's (2014) third-generation expansion broadened the framework further to account for interacting systems and systemic contradictions. Engeström's model (see Figure 1) describes an activity system in which the *Subject* consists of individuals or groups working towards a common objective.

Within an activity system, the *Subject* refers to the individual(s) or collective(s) engaged in the activity (e.g. academics teaching engineering students); *Tools* encompass the physical or conceptual instruments employed to mediate the object (e.g. specialist software, professional standards, and assessment frameworks); *Object* denotes the goal or purpose motivating the activity system (e.g. producing graduates who are able to apply discipline-specific digital tools in professional contexts); *Rules* comprise the explicit or implicit policies and cultural conventions that govern activity (e.g. university policies, accreditation requirements, or student expectations); *Community* encompasses

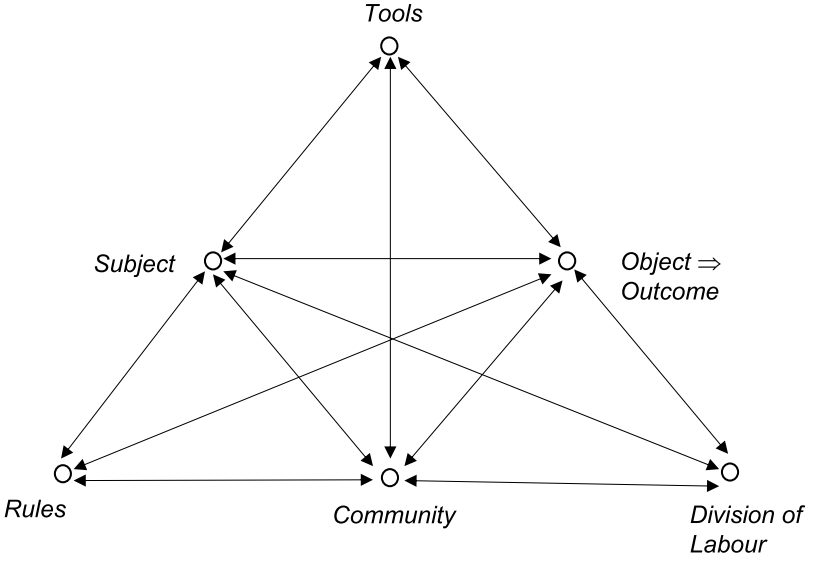


Figure 1. Activity system model (based on Engeström, 2014).

those participating in the activity alongside the context in which it takes place (e.g. academics, students, and professional services staff); *Division of Labour* describes how responsibilities are apportioned among system members (e.g. departmental hierarchy and workload distribution); and *Outcome* designates the intended and unintended consequences arising from the activity (e.g. graduates who are able to apply discipline-specific digital tools in professional contexts, or, unintentionally, students who focus strategically on assessment requirements rather than broader capability development).

Selection and justification of Activity Theory

Teaching is a complex activity requiring a theoretical framework that accounts for multiple participants, instruments, conventions, and power dynamics. Alternative frameworks were considered but found to be insufficient for this study's purposes. The Technology Acceptance Model (Davis 1989) and TPACK (Mishra and Koehler 2006) focus primarily on technology adoption or individual competencies, lacking Activity Theory's capacity to analyse systemic tensions. Community of Practice (Wenger 1998) examines shared learning but offers limited insight into contradictions that drive transformation.

Activity Theory was selected as the most appropriate framework for this study for three key reasons. First, it provides a structured approach for exploring the multifaceted activity of academics developing students' digital capabilities against a backdrop of workload intensification. Unlike frameworks that focus primarily on individual behaviour or technological determinism, Activity Theory examines how *Tools*, *Rules*, *Community*, and *Division of Labour* interact to shape teaching practices.

Second, Activity Theory has demonstrated utility in HE research contexts similar to this study. Bligh & Flood's (2017) analysis of 59 empirical research papers demonstrates Activity Theory's established value in HE research. Specific applications include conceptualising postgraduate student research (Hammond 2019), understanding interactions between interdisciplinary teams (McCance et al. 2023), and exploring unbundling of the HE curriculum (Cliff et al. 2020). Park and Jo (2016) employed Activity Theory to identify barriers in technology integration, whilst Grönlund, Samuelsson, and Samuelsson (2021) explored tensions in blended learning environments – both studies addressing questions about contradictions in educational activity systems that are directly relevant to the present work.

Third, Activity Theory's emphasis on contradictions and tensions makes it suitable for examining the challenges faced by academics attempting to develop students' digital capabilities amidst competing demands and limited resources, aligning directly with the research questions.

Analytical focus on contradictions

A fundamental part of an activity system is its contradictions, which are structural tensions that have progressively accumulated and can be catalysts for development, learning or change. Engeström (2014) describes four levels of systemic contradiction:

- *Primary contradictions* manifest within a single node of the activity triangle (e.g. tensions within the *Tools* node, such as software that simultaneously enables and constrains learning).
- *Secondary contradictions* arise between elements of an activity system in ways that may undermine each other (e.g. when assessment *Rules* conflict with the *Object* of developing digital capabilities).
- *Tertiary contradictions* occur between different versions of the same activity system (e.g. tensions between existing and emerging models of teaching digital capabilities).
- *Quaternary contradictions* emerge between the activity system and those neighbouring it (e.g. tensions between the academics' teaching activity system and the management activity system).

This framework provides specific analytical perspectives through which to examine the data collected from participants, enabling systematic identification of barriers and enablers to effective digital capabilities development in HE.

Research design

Case study approach

A case study approach has been selected as it facilitates contextualised in-depth and nuanced investigations of complex issues. The case is bounded by role (academics), organisation (a UK university), process (digital capabilities development), and period (current practice) (Miles, Huberman, and Saldaña 2020). This methodological choice is appropriate for examining the multifaceted interactions between academics, digital tools, and institutional contexts that characterise digital capabilities development.

Research site and context

The research site is a research-intensive HE institution in the UK that is highly ranked for engineering (Complete University Guide 2023; The Guardian 2023; Times Higher Education 2023). The institution has invested significantly in digital capabilities development, including subscription to a self-assessment platform and online learning resources. Activity within the Faculty of Engineering and Physical Sciences (FEPS) provides an appropriate context for examining digital capabilities development by academics at research-intensive universities.

Positionality statement

As a constructivist-interpretivist, my research stance is shaped by Lincoln & Guba's (2016) exploration of the dual nature of reality encompassing its physical manifestation alongside the web of associated meanings and social interactions – which in practice means that participants' accounts are treated as interpretations of experience rather than objective reports of practice. Central to this perspective is the acknowledgment of research participants as co-creators of knowledge, with their viewpoints accorded equal significance.

I have been employed at the research institution for over a decade, establishing a network of contacts and developing a deep understanding of the institution's values and customs. This cultural familiarity provides a common frame of reference with participants and facilitates drawing inferences from their responses. As Hampton, Reeping, and Ozkan (2021) note, researcher positionality influences each phase of the research process, from initial construction of research questions through to dissemination of findings.

This constructivist-interpretivist stance informed both methodological choices and interpretive processes throughout the study, prioritising participants' meanings and experiences. My position as an institutional insider brings both advantages and methodological challenges. This insider status enables nuanced understanding of contextual factors and facilitates participant recruitment and rapport-building. However, my employment within an IT-related department might have led some participants to perceive an agenda to promote technology use. To address this concern, participants were reassured at the outset of each interview that responses would remain anonymous and that the research was not evaluative in nature; this was also stated explicitly in the participant information sheet.

I approached this work from an interpretive paradigm, acknowledging that analysis is inherently shaped by personal experiences and perspectives. My background in learning design and technology implementation influenced the interpretation of participants' responses regarding digital tools and capabilities. This background offered valuable insights for understanding nuances in

participants' accounts whilst potentially introducing biases towards technology-oriented solutions. Equally, deep institutional familiarity carries a risk of normalising practices that an outsider might question; I sought to guard against this by treating familiar institutional arrangements as analytically significant rather than assumed. Throughout the research process, I maintained reflexivity about these influences through regular journaling, whilst recognising that complete objectivity is neither possible nor desirable in interpretive research. This approach aligns with what Hampton, Reeping, and Ozkan (2021) describe as making 'the invisible decisions and interpretations of the researcher visible in the study'.

My positionality influenced data collection and analysis processes. During interviews, shared institutional knowledge created opportunities for deeper probing of responses based on contextual understanding. In the analytical phase, my familiarity with institutional systems and practices helped identify tensions between stated policies and lived experiences of participants. This stance influenced how contradictions within activity systems were identified and interpreted, with particular attention to how participants made sense of institutional tensions. My commitment to constructivist principles shaped decisions to validate interpretations through member checking – whereby participants were provided with summaries of key analytical interpretations and invited to comment, with any resulting clarifications incorporated into the final analysis – and to present findings in ways that preserved participants' voices alongside analytical insights.

Sampling and participants

Participants were purposively sampled to include academics whose roles position them to reveal the tensions involved in integrating digital capabilities into engineering teaching. Education-focused academics were selected because they carry the primary burden of curriculum delivery and are therefore ideally positioned to illuminate the challenges of implementing new curriculum requirements within the context of workload intensification. These academics typically have substantially heavier teaching loads than other academic roles, engage more intensively with curriculum design, and focus specifically on student learning outcomes (Bull et al. 2024). Their experiences reveal systemic tensions that affect all academics but are acute and visible in this group due to their concentration in intensive instructional duties.

Participants were recruited through convenience sampling via email and snowballing, targeting education-focused academics from FEPS who had taught at the institution for at least two years. Six academics teaching engineering students were recruited, representing a range of levels from foundation year (for high-achieving students without the necessary pre-requisites) to master's degree. To maintain confidentiality, pseudonyms are used: Alex, Boris, Chris, Della, Eric, and Felicity. The sample included two academics involved with the foundation year programme, three academics teaching core engineering modules, and one academic with both teaching and programme leadership responsibilities.

The sample size aligns with established qualitative research principles for case study methodology. Malterud, Siersma, and Guassora (2015) propose the concept of 'information power', suggesting that the more relevant information a sample holds for the study's aims, the fewer participants are needed. Guest, Bunce, and Johnson (2006) found that data saturation often occurs within the first twelve interviews, with basic thematic elements present after just six interviews. The current sample achieved sufficient depth through purposeful selection of participants representing various teaching roles, allowing for theoretical saturation of key concepts under investigation (Hennink, Kaiser, and Marconi 2017).

The focus on a single faculty within one institution allows for contextual coherence and analytical depth, though it necessarily limits wider applicability. This study aims for theoretical generalisability through detailed analysis of patterns and mechanisms that may be transferable to similar contexts. This approach aligns with case study methodology's emphasis on depth rather than breadth

(Flyvbjerg 2006) and is consistent with Stake's (1995) emphasis on particularisation rather than generalisation as the primary strength of case study research.

Data collection method

Six semi-structured online interviews of ~60 min were conducted in early 2024 using Microsoft Teams. The interview guide drew on Mwanza-Simwami's (2011) AODM questions, which were specifically developed to operationalise Activity Theory in research contexts.

The interview guide (available in Appendix A) included questions such as: 'Are there any institutional policies or guidance, or disciplinary or professional norms that influence your approach to developing student digital capabilities?' Questions were organised around key Activity Theory components (*Subject*, *Tools*, *Object*, *Rules*, *Community*, *Division of Labour*) to ensure comprehensive coverage of the activity system.

Not all questions were asked in every interview; rather, the semi-structured approach allowed for following participant responses and exploring emergent themes whilst ensuring coverage of core Activity Theory components. Interviews explored participants' perceptions of their role and workload, views on curriculum expectations, and perspectives on student digital capabilities.

Data management

Ethics approval was gained from the institutional research ethics committee. Participants provided informed consent and had the opportunity to review their interview transcripts and request redactions or clarifications. All interviews were recorded with permission and transcribed verbatim.

Data analysis

Deductive coding of anonymised interview transcripts was conducted using NVivo software, employing a codebook derived from Activity Theory (Engeström, 2014). The codebook (summarised in Appendix B) included operational definitions for each Activity Theory component and contradiction level, along with example indicators.

The analysis process involved six stages: familiarisation with interview data through reading and re-reading transcripts; initial coding using the Activity Theory-based codebook; identification of contradictions at primary, secondary, tertiary, and quaternary levels; theme development through grouping related codes; review and refinement of themes against coded extracts; and final analysis interpreting findings through the Activity Theory perspective.

Participants' contributions were mapped onto activity system models to model specific situations and explore contradictions. The resulting models surfaced the most important contradictions that are discussed in the findings section.

Findings

This study's findings address how academics engage with student digital capabilities, and the contradictions academics experience when attempting to integrate digital capabilities into their teaching activities. Participants' experiences arise within the context of their practice: teaching within an engineering faculty (see Figure 2).

The findings are presented in two parts. First, I describe the key components of the academics' activity system (*Subject*, *Tools*, *Object*, *Community*, *Rules*, and *Division of Labour*) as represented in Figure 2, establishing the context within which digital capabilities development occurs. Second, I examine the contradictions within this activity system, and between it and neighbouring systems, focusing on secondary contradictions (between elements within the academics' system) and

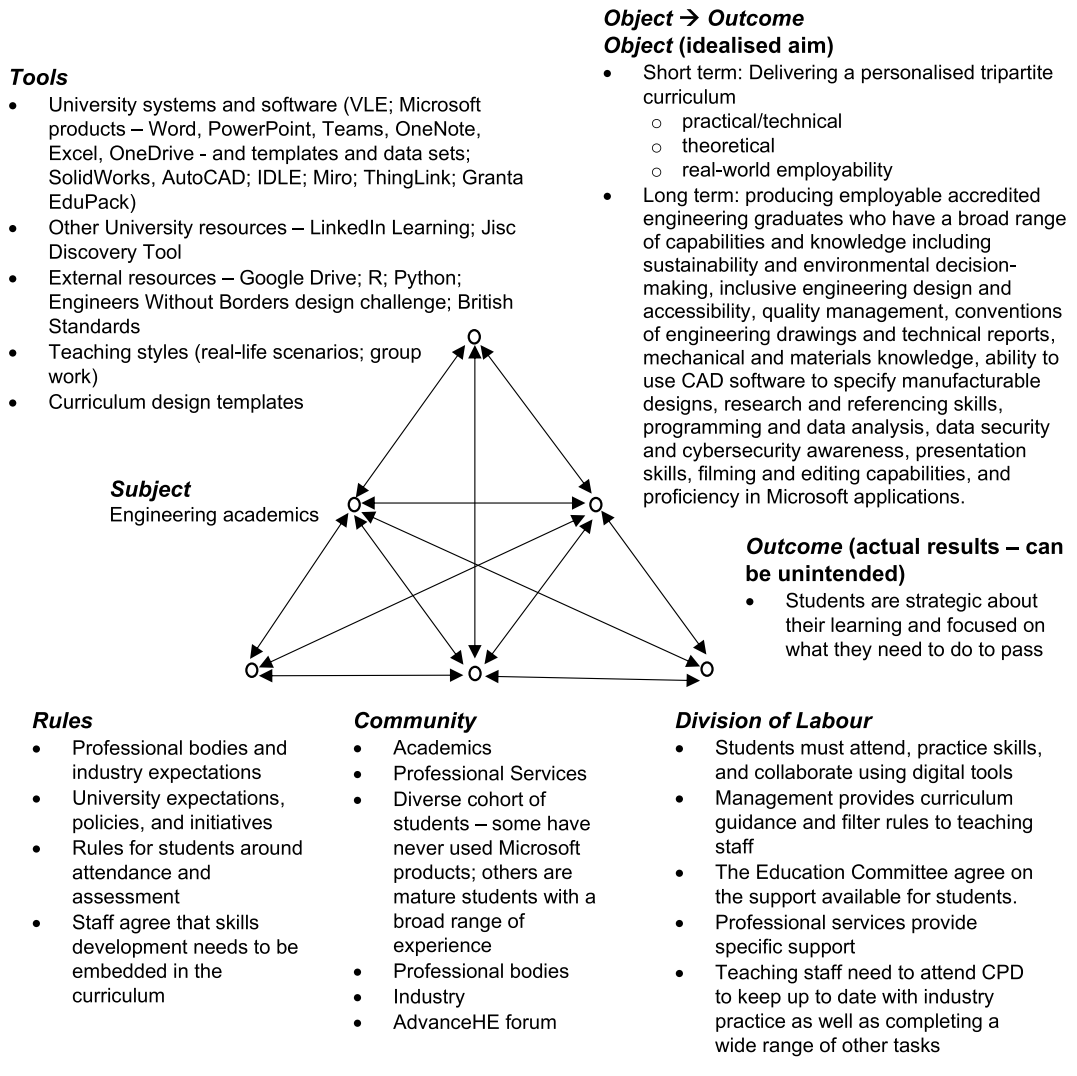


Figure 2. An activity system model of academics teaching engineering students.

quaternary contradictions (between the academics' system and the management system). These contradictions emerged as the most significant barriers to effective digital capabilities development.

The academics' activity system

Figure 2 illustrates the complete activity system of academics teaching engineering students. Whilst the figure provides a visual overview of all components and their relationships, the following sections elaborate on each element, highlighting aspects relevant to digital capabilities development and the tensions that emerged in practice. It is possible for contradictions between any two nodes on the system to emerge, even if they are not directly adjacent, such as *Tools* and *Division of Labour*.

Subject

Activity Theory postulates that human activities are collaborative endeavours governed by *Objects* and mediated by *Tools* (Bligh and Flood 2017). In this activity system, the *Subjects* are academics who teach engineering students. Their activities encompass lecturing, content creation, teaching

software use, assessment, and research alongside substantial administrative responsibilities including committee work and personal tutoring. Alex described them as 'a diverse group of people who have all sorts of different interests working in different places', reflecting the varied expertise within engineering academics.

Tools

The *Subjects* used various *Tools* to support teaching. These included both digital learning environments and discipline-specific technologies. Felicity, Eric, Della, and Chris discussed teaching methodologies like using real-life scenarios and group work to develop students' employability. Participants used Microsoft software and technical templates to model good practice. Specialist engineering software – including CAD programs, programming environments (Python, R), and materials analysis tools (Granta EduPack) – formed essential components of the technical curriculum. Awareness and utilisation of institutional digital capability tools was limited primarily to foundation year programmes.

Object ◇ outcome

The long-term *Object* of the academics' activity was to produce employable engineering graduates who have a broad range of capabilities. However, participants also identified a short-term *Object*: delivering a personalised tripartite curriculum that balances practical/technical skills with theoretical knowledge and real-world employability. Participants endeavoured to make their students successful at university and in work by embedding digital literacy skills in everyday work: 'It's always been about, 'OK, how useful is this person actually going to be in the real world?' And that's always been a real focus of what I've done'. [Eric]

Much of the breadth of capabilities and knowledge that participants said graduates are expected to develop is tacit, and not directly referenced in module-level learning outcomes:

> We kind of expect the ability to use the software as a means to achieving a broader end within engineering. Similarly, we ask people to make presentations, and we often enjoy their output because they've been able to illustrate it beautifully, set it out in a PowerPoint or equivalent slide format, and illustrate it with video … It's not the only way to make a presentation, but we kind of assume it is. [Alex]

Outcome refers to the intended or unintended implications that result from the activity. A positive *Outcome* is that students achieve the module-level learning outcomes and can overcome challenges; however, they are strategic and only focus on passing each module rather than developing broader capabilities.

Community

Community encompasses stakeholders who influence but do not set the agenda of the activity, including academics, students, professional services staff, and external bodies. The engineering education environment involves diverse groups with varied expertise and perspectives.

Professional services staff play important supporting roles. Chris described collaboration with specialist skills and digital learning teams, and access to institutional education networks. External stakeholders such as industrial advisory boards and professional bodies also shape the teaching environment.

The student community is characterised by diversity. Participants consistently noted variation in students' prior experience and digital capabilities. This heterogeneity affects how teaching activities are designed and delivered, with academics needing to accommodate different starting points and learning needs within their cohorts.

Rules

Rules influencing this activity system are both explicit and implicit. Participants focused on meeting EngC's fourth edition of Accreditation of Higher Education Programmes (AHEP4), internal quality assurance (QA) processes, and incorporating sustainability requirements.

Participants stated that academic skills are expected to be embedded within practice; otherwise, students disengage. Some mentioned needing to be digitally capable, maintain current disciplinary knowledge, and manage their workload alongside other expectations. An implicit *Rule* was that academics should closely follow policies set by management, though most participants agreed there was freedom to teach within their specialism.

Student behaviour is governed by parallel *Rules*. There are tacit expectations around students' attendance at timetabled sessions, and collaboration on group projects using digital tools. An implicit *Rule* is that students must develop 'graduate capabilities', which are not explicitly referenced in module-level learning outcomes.

Division of labour

Labour is divided vertically by authority (Associate Dean Education; Deputy Head of School Education; Programme Leaders) and horizontally by specialty (internal QA team; Education Services staff; librarians). However, participants expressed uncertainty about how some responsibilities are distributed. Alex explained: 'Many of our staff are directed towards their various responsibilities and haven't been directed towards being responsive to these types of questions' [around software usage].

Eric described a disconnect between his expertise and his workload: 'I don't know why I got a design degree … What I actually need is an admin degree because I spend all my time filling in spreadsheets and doing Word documents'.

The FEPS Education Committee decides on the *Division of Labour* and communicates teaching requirements, filtering *Rules* to academics. The internal QA team interprets requirements such as subject benchmark statements from the Quality Assurance Agency for Higher Education (QAA 2023) – the UK's independent quality body that develops discipline-specific standards similar to accreditation requirements in many countries – to ensure these national quality standards are being met. Education Services staff create guides to using software, develop learning playlists and create video tutorials. Librarians deliver sessions on research skills when invited.

Contradictions affecting digital capabilities development

Analysis identified all four levels of contradictions proposed by Engeström (2014). However, given the constraints of article length and to provide sufficient depth of analysis, this paper focuses on secondary contradictions operating between elements within the academics' activity system (see Figure 3), and quaternary contradictions occurring predominantly between the academics' activity system and the management activity system (see Figure 4 and Table 1). The selected contradictions emerged as the most salient barriers to effective practice during analysis.

Secondary contradictions within the academics' activity system

Secondary contradictions – tensions between different elements within a single activity system – manifested in multiple ways that directly impeded digital capabilities development. The following subsections examine the most significant of these contradictions.

Secondary contradictions between resources for supporting students (Tools) and producing capable graduates (Object)

Multiple secondary contradictions emerged between the *Tools* available for supporting students and the *Object* of developing students' digital capabilities. Despite expectations that students are

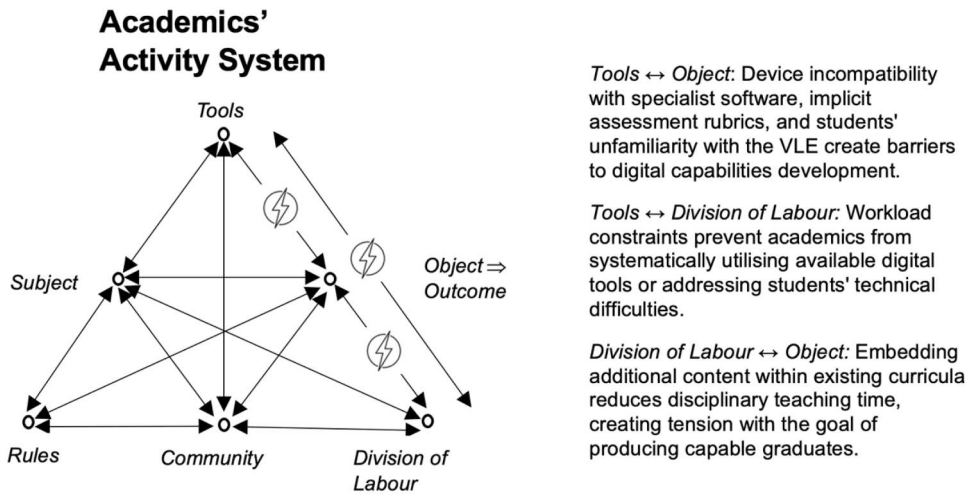


Figure 3. Activity system model showing secondary contradictions within the system (indicated by lightning bolts between elements). Adapted from Engeström (2014).

digitally capable, the virtual learning environment (VLE) created unexpected barriers. Eric noticed 'a strange disconnect' in how different generations use digital media, observing that students 'can't quite get their heads around VLEs and email to the same degree that we can'. Most participants now provide VLE orientation sessions, adding to their teaching responsibilities. Further *Tools* ↔ *Object* contradictions emerged from incompatible personal devices, and unclear rubrics. Students'

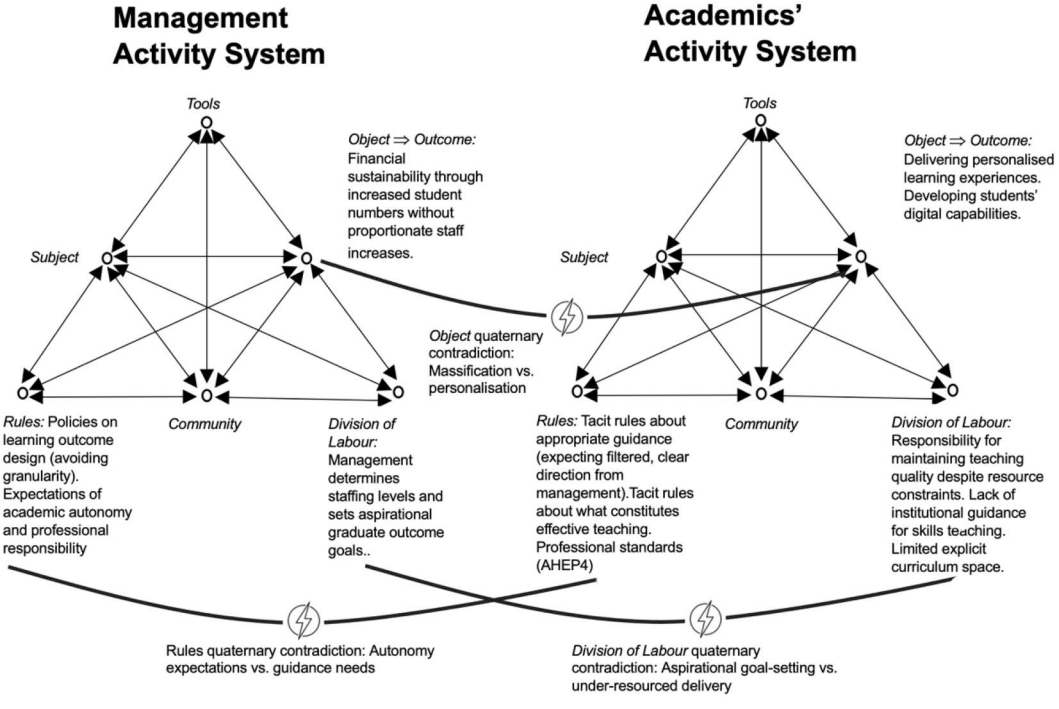


Figure 4. Activity system model showing quaternary contradictions (lightning bolts) between the academics' activity system and the management activity system. Adapted from Engeström (2014).

Table 1. Activity system elements and quaternary contradictions.

System nodes	Management activity system	Academics' activity system	Contradictions
Subject	• Senior managers • Institutional leadership	• Academics	
Tools	• Policies and procedures • Metrics and KPIs • Strategic planning documents • Resource allocation models	• VLE (Blackboard) • Specialist software (CAD, Python, R) • Assessment rubrics • Teaching methods and materials	
Object ⇒ Outcome	• Financial sustainability through increased student numbers (without proportionate staffing increases) • Producing 'globally recognisable' graduates	• Delivering personalised learning experiences • Developing specific, demonstrable digital capabilities in students	• Massification vs. personalisation • Management's pursuit of financial sustainability through increased student numbers (massification) creates tensions with academics' *Object* of delivering personalised learning experiences.
Rules	• Policies on student-to-staff ratios • Resource allocation policies • Learning outcome design *Rules* (avoiding granularity) • Academic autonomy and professional responsibility expectations	• Tacit *Rules* about appropriate guidance (expecting filtered, clear direction) • Tacit *Rules* about what constitutes effective teaching • Professional standards (AHEP4)	• Autonomy vs. guidance needs • Management *Rules* about academic autonomy and learning outcome design conflict with academics' tacit *Rules* about appropriate guidance and the need for explicit skill development.
Community	• Senior leadership • Institutional stakeholders • Governing bodies	• Academic colleagues • Students • Professional services staff	
Division of Labour	• Management determines staffing levels • Set aspirational graduate outcome goals • No central mechanism for organising skills teaching	• Responsibility for maintaining teaching quality despite resource constraints • Lack of institutional guidance for skills teaching • Limited explicit curriculum space	• Aspirational goal-setting vs. under-resourced delivery • Management's *Division of Labour* decisions (determining staffing levels without implementation mechanisms) create structural misalignment with academics' responsibility for maintaining teaching quality with impossible resource constraints.

personal devices often lack compatibility with specialist software, whilst assessment expectations about digital capabilities and presentation skills remain implicit rather than explicit in rubrics. Felicity noted, however, that detailed rubrics with explicit criteria can reduce this tension.

A related secondary contradiction existed between *Tools* and the *Division of Labour*: whilst digital tools were available to support teaching, the distribution of teaching responsibilities meant that academics had insufficient time to fully utilise these resources or address the technical difficulties students encountered – with Felicity describing frustration with 'dealing with the hardware rather than the learning' when substantial disciplinary material required coverage.

Secondary contradiction between time constraints (Division of labour) and producing capable graduates (Object)

Participants agreed that embedding additional content in curricula is the most effective delivery method, but a secondary contradiction exists because it reduces disciplinary teaching time. Alex explained: 'I can't do too much sustainability because then I've got to really spend time introducing what sustainability means and that impacts on how I do that in my class and what I make my focus on'.

Boris tried extracting co-curricular topics for personal academic tutors to deliver: 'Our personal academic tutoring … That takes a lot of content … So, it is not entangled in there in our academic

modules … even if they are not part of the learning objectives of a specific module, they are addressed'.

Quaternary contradictions among activity systems

Quaternary contradictions – tensions between the academics' activity system and neighbouring systems – proved significant because they represent structural misalignments that cannot be resolved through individual pedagogical innovations. Three major quaternary contradictions emerged between the academics' activity system and the management activity system.

Quaternary contradictions between management aspirations and academics' implementation

A quaternary contradiction arose between the *Object* and *Rules* of the management activity system and those of the academics' activity system. Management's *Object* of producing 'globally recognisable' graduates (University of Southampton 2024) exists at a strategic level of abstraction that creates tensions with academics' *Object* of developing specific, demonstrable capabilities in students.

This contradiction is exacerbated by management's *Rules* about learning outcome design, which intentionally avoid the level of granularity that would make skill development explicit and assessable. Boris noted that 'from year 2 there are no specific mentions of digital capabilities … so, it's an expectation, but there is no specific training addressing it'.

The *Division of Labour* further compounds this quaternary contradiction: whilst management sets aspirational goals, there is no central mechanism for organising the skills teaching that would be required to achieve these goals. Academics expect students to have specific skills but lack both the institutional guidance and the explicit curriculum space to teach them systematically.

Quaternary contradictions between autonomy expectations and guidance needs

A quaternary contradiction arose between the *Rules* and *Division of Labour* in the academics' activity system and those in the management activity system. Academics' tacit *Rules* about appropriate guidance (expecting filtered, clear direction) conflicted with management's *Rules* about academic autonomy and professional responsibility. This manifested in tensions around the *Division of Labour*: academics expected management to interpret and filter external requirements, while management appeared to expect academics to engage directly with regulatory documents themselves.

Academics felt that management communications neither shared enough information nor made expectations clear. This tension made participants feel liberated, but directionless. Most stated that important information is not directly communicated to them as they lack time to read through regulations. Eric hoped that 'those people [management] are keeping an eye on those particular guidelines and filtering those down to me'.

This information filtering meant participants were unfamiliar with QAA subject benchmark statements and whether they align with EngC's AHEP4. Some emphasised that a clearer understanding of requirements and more direction regarding teaching content and resources would be beneficial:

> I don't really have anyone saying to me, 'You need to make sure you do it. They're doing this. But I think we could do that bit'. But I think that needs to be a bit more better in the school. It would help me if some people would say, ' … they need to be able to do XYZ'. [Chris]

> You know, I actually want somebody to be looking at that stuff, saying, 'OK, you're doing it right. You're doing it wrong. You should need to change this. You need to do this'. But that seems to be pretty much absent, which is a new experience for me. [Eric]

This was echoed by Felicity, who described receiving feedback on her teaching and content in previous institutions.

Quaternary contradictions between personalisation and massification

A quaternary contradiction arose between the *Objects* and *Rules* of the academics' activity system and those of the management activity system. The academics' *Object* of delivering personalised learning experiences – supported by their tacit *Rules* about what constitutes effective teaching – conflicted with management's *Object* of financial sustainability achieved through increased student numbers without proportionate increases in staffing. This manifested in the *Division of Labour*, where academics bore responsibility for maintaining teaching quality despite resource constraints imposed by management decisions.

Eric, Della, and Felicity spoke extensively about the challenges of teaching increasing cohort sizes. Della described managing 1,000 students in year one and year two, and Felicity described her course size nearly tripling in eight years. Eric described the challenge: 'How do you effectively teach a cohort of nearly 600 students? […] trying to think of new methods to give what feels like a valuable one-to-one experience'.

Most participants stated that staffing levels were challenging in terms of providing appropriate student support. Participants felt that in-person training could be more beneficial, but students are directed to eLearning because attendance at requested workshops has been low.

Emerging changes and limitations of the dataset

The limited discussion of GenAI in the data represents a notable consideration for interpreting the findings. Only one participant, Felicity, explicitly mentioned AI's impact, noting that ChatGPT 'really changed the assessment structure,' leading her to shift to 'a coursework and a viva to be able to actually assess that that is their work'. This singular reference, despite interviews being conducted in early 2024, may reflect the emerging nature of institutional responses to GenAI technologies at that time.

Emerging changes include employment of a CAD specialist and exploration of new ways of delivering content online with targeted workshops for students needing additional help.

Synthesis

The findings reveal interconnected contradictions operating at multiple levels within the academics' activity system, and between it and the neighbouring management activity system. Secondary contradictions between *Tools*, *Objects*, and *Division of Labour* highlight how resource constraints, unclear expectations, and time pressures impede digital capabilities development within academics' immediate practice. Quaternary contradictions between academic and management activity systems expose deeper structural tensions around guidance, autonomy, and resource allocation that cannot be resolved through individual pedagogical adaptations alone. Collectively, these contradictions illuminate why digital capabilities integration remains challenging despite widespread recognition of its importance, pointing to the need for systemic rather than individual-level solutions.

Discussion

This study investigated how academics engage with student digital capabilities in their teaching, focusing on the instruments they utilise and the tensions they experience. The findings both align with and extend previous research on digital capabilities in HE whilst revealing unexpected insights about academic autonomy and guidance-seeking behaviours. The discussion addresses each research question in turn, with particular attention to an unexpected finding regarding academics' desire for clearer managerial guidance – a finding that challenges traditional notions of academic freedom prevalent in the literature.

RQ1: How do academics engage with student digital capabilities in their teaching?

Findings from this research confirm that academics recognise the importance of developing students' digital capabilities for employability, aligning with previous research (Lewis 2023; Van Essen-Fishman, Morgan, and Madrigal 2023). However, whilst this is a priority in the literature, the present study reveals a gap between recognition and implementation. Participants incorporated skills development into their teaching practices, but primarily within the constraints of existing curricula and workloads, rather than through comprehensive redesign of teaching activities.

Making implicit expectations explicit through assessment design

Participants identified the need for more detailed rubrics and module-level learning outcomes to produce digitally capable graduates. This finding supports previous research on the importance of aligning digital literacy with broader educational objectives (Austen et al. 2016) whilst expanding upon it by specifically highlighting assessment design as a key mechanism for embedding digital capabilities. The literature emphasised curriculum design importance, but findings suggest that assessment criteria may be equally important in driving student engagement with digital skills development.

The gap between implicit expectations and explicit assessment emerged as a significant barrier with transferability beyond engineering education. The principle of making implicit expectations explicit through detailed assessment criteria applies across contexts where students develop capabilities transcending traditional disciplinary boundaries.

Assessment rubrics must explicitly address both disciplinary knowledge and digital capabilities. Explicit assessment criteria can reduce the contradiction between *Tools* (rubrics) and *Object* (capable graduates).

Pragmatic adaptation to massification pressures

The findings confirm previous research on the challenges of delivering personalised learning experiences in the context of increasing student numbers (Hartman and Darab 2012). However, the present study extends this understanding by highlighting how institutional investment in digital tools may remain underutilised due to limited awareness. This aligns with Gregory and Lodge's (2015) observations about discrepancies between institutional investment in digital resources and their effective implementation whilst adding nuance by identifying specific barriers to tool adoption within engineering disciplines.

Participants demonstrated pragmatic adaptations to massification pressures, such as using recorded lectures and software tutorials for large cohorts. These strategies address teaching at scale whilst creating opportunities for students to develop digital capabilities through self-paced engagement with technical content, though they place the burden of innovation on individual staff rather than addressing systemic under-resourcing. Supporting digital capabilities development requires adequate staffing levels and resources.

RQ2: What tensions and contradictions do academics experience when attempting to integrate digital capabilities within their teaching activities?

Academics experience several significant tensions and contradictions when attempting to integrate digital capabilities within their teaching activities, as revealed through the interviews and subsequent Activity Theory analysis. The study identified both secondary contradictions (within the academics' activity system) and quaternary contradictions (between the academics' activity system and neighbouring systems, particularly management). These contradictions illuminate systemic barriers to effective digital capabilities development that extend beyond individual pedagogical choices.

The guidance paradox: autonomy versus support needs

A significant finding was that academics expect clearer guidance on expectations from managers. This contrasts with much of the existing literature on academic autonomy, which often emphasises academics' desire for independence in curriculum design and teaching methods (Dowling-Hetherington 2014; Vostal 2015). Whilst previous studies have focused on the loss of autonomy as a negative consequence of increased managerial control (Kinman and Jones 2003), participants in this study expressed a need for clearer direction, particularly regarding digital capabilities integration.

This unexpected finding suggests that in contexts of workload intensification and competing priorities, academics may prioritise efficiency and clarity over complete autonomy, representing a potential shift in how academic freedom is conceptualised under current pressures. This finding extends existing understanding by highlighting how context shapes academic preferences regarding autonomy versus guidance. The literature on academic workload intensification has documented loss of autonomy as problematic (Dowling-Hetherington 2014; Kinman and Jones 2003), but this study reveals a more nuanced picture: academics may actively seek structured support when facing multiple, competing demands.

The transferability of this finding warrants careful consideration. The guidance-seeking behaviour may be most applicable in contexts where: (1) academics face workload intensification alongside multiple curriculum requirements; (2) there are complex external regulatory frameworks (such as professional accreditation requirements); and (3) institutional structures create distance between strategic decision-making and teaching delivery.

Structural tensions of workload intensification and massification

The study corroborates extensive literature on workload intensification and the massification of HE (Hartman and Darab 2012; Kinman and Jones 2003), demonstrating how these factors continue to impact academics' capacity to engage with redesigning teaching activities. The present study extends this research by drawing explicit connections between these broader institutional pressures and specific challenges in digital capabilities integration.

Participants noted that larger and more diverse cohorts create additional tensions. They discussed the benefits and contradictions of having a foundation year – these students develop the digital competencies required for students in HE but may not have such in-depth subject knowledge as other direct entrants, creating tensions in teaching approach across different student groups.

Some participants experienced tensions between the need for their own professional development in digital tools and pedagogical innovation, and the reality of limited time and resources. They described a tension between their responsibilities as educators and student autonomy, feeling that students needed to take more responsibility for their own development by responding to feedback.

This study's contribution lies in explicitly mapping connections between the massification of HE, workload intensification, and development of students' digital capabilities. The findings suggest that a holistic approach is needed, one that integrates clear institutional strategies, robust support mechanisms, and enhanced resource allocation whilst acknowledging the complex interplay between workload pressures, teaching expectations, and student needs.

Synthesising the contradictions: systemic barriers to digital capabilities development

The various contradictions identified – from guidance-seeking behaviours to workload pressures to assessment clarity – operate as interconnected elements within a complex activity system. The quaternary contradictions between the academics' activity system and the management activity system are significant, as they reveal fundamental misalignments in *Object*, *Rules*, and *Division of Labour* that cannot be resolved through individual pedagogical adaptations alone. Effective digital capabilities integration requires systemic change addressing institutional structures, resource allocation, and

communication mechanisms, rather than placing responsibility solely on individual academics to innovate within increasingly constrained conditions.

Transferability of findings

Whilst this study focused on engineering education at a single UK university, the findings have transferability to other contexts through several mechanisms, though the extent of transferability varies depending on whether the contextual features are common to curriculum change generally or specific to digital capabilities development. The identification of tensions between workload intensification and curriculum development likely applies across disciplines facing similar massification pressures, particularly those with technical or professional components requiring regular updates to remain current.

The contradictions between management expectations and academic implementation illuminate power dynamics that transcend disciplinary boundaries, suggesting that the communication patterns identified here may operate similarly in other faculties and institutions. The methodology employed – using Activity Theory to analyse contradictions within teaching systems – provides a portable analytical approach that researchers could apply in diverse educational settings to uncover similar tensions.

The findings may be particularly transferable to other STEM disciplines where digital capabilities are similarly embedded within disciplinary practice rather than taught as standalone skills. The identified need for detailed assessment rubrics that make implicit expectations explicit represents a principle that could be productively applied across various educational contexts, regardless of discipline or institutional setting.

The findings are most directly applicable to contexts with similar structural conditions: research-intensive institutions with academics carrying substantial teaching loads, disciplines with professional accreditation requirements, and systems experiencing massification pressures.

It is important to acknowledge that some contradictions identified – particularly those relating to workload intensification, communication between management and academics, and assessment clarity – could arise during implementation of any substantial curriculum change, not solely digital capabilities integration. However, what distinguishes digital capabilities integration are several factors: the technical specificity of required tools and competencies that cannot simply be 'added' to existing content; the rapid pace of technological change requiring continuous updating rather than one-off curriculum revisions; the intersection with professional accreditation requirements that mandate specific digital competencies; and the assumption (often incorrect) that students arrive with foundational digital skills, which differs from other curriculum additions where prerequisite knowledge gaps are more readily acknowledged. The transferability of findings should therefore be understood as strongest in contexts where curriculum changes share these characteristics – technical complexity, mandatory external requirements, and contested assumptions about student preparedness.

What makes certain findings specific to engineering education is the technical specificity of required digital capabilities – such as CAD proficiency, computational methods, and simulation techniques – which are fundamental to professional practice rather than merely supplementary skills as they might be in other disciplines. The extensive requirements from engineering accreditation bodies like the Engineering Council and International Engineering Alliance create additional pressures specific to engineering programmes. Engineering's characteristically large cohort sizes and the hands-on nature of digital capability development in technical disciplines create a challenging environment for digital capability integration.

Practical recommendations

The following recommendations emerge from the integration of this study's findings with existing literature on digital capabilities development and workload management in higher education. Each

recommendation addresses specific contradictions identified in the activity system analysis whilst acknowledging the resource constraints under which academics operate.

For institutional leadership

- **Implement structured information filtering mechanisms** where senior academics review external requirements and provide simplified, discipline-specific guidance to teaching staff. This recommendation directly addresses the quaternary contradiction between the academics' activity system and the management activity system that was explicitly articulated by participants.
- **Develop clear templates for assessment rubrics** that explicitly include digital capabilities components, ensuring consistent assessment across modules. This recommendation emerges from the secondary contradiction identified between assessment tools and the object of producing digitally capable graduates. Participants noted that expectations often remain implicit, which contrasted with one participant's explicit rubrics, which demonstrated a successful approach to making digital capability expectations transparent. This aligns with literature emphasising the importance of aligning assessment of digital capabilities with learning objectives (Austen et al. 2016).
- **Establish dedicated time allowances within workload models** specifically for digital capabilities development activities. This recommendation addresses the pervasive workload intensification documented throughout the findings. Participants described having 'very little time for reflection in any way shape or form, let alone for professional development', and characterised their roles as spending time on administrative tasks rather than their areas of expertise.

For programme leaders

- **Establish clear communication channels** about digital capability expectations across year groups to address the contradiction between implicit and explicit skill development. Programme leaders could coordinate explicit mapping of where digital capabilities are taught, practised, and assessed, ensuring students understand expectations throughout their studies.
- **Coordinate with professional services** to develop discipline-specific digital capability support that reduces burden on teaching staff. The findings revealed limited awareness of existing institutional tools beyond foundation year programmes.

For individual academics

- **Adopt minimal viable redesign approaches**, identifying one digital capability to explicitly embed per teaching cycle rather than attempting comprehensive curriculum redesign. This recommendation directly addresses the secondary contradiction between time constraints and producing capable graduates. The minimal viable redesign approach acknowledges the temporal compression documented by Hartman and Darab (2012) and Vostal (2015), providing a pragmatic strategy for incremental improvement. Drawing on my positionality as an institutional insider familiar with competing curriculum demands, this recommendation reflects the reality that sustainable change often occurs through small, manageable adjustments rather than wholesale transformation.
- **Develop detailed assessment rubrics** that explicitly address both disciplinary knowledge and digital capabilities, making implicit expectations explicit.
- **Utilise existing institutional tools at strategic points** in the curriculum where students would benefit most from self-assessment and reflection.

- **Engage with the scalability strategies** that colleagues have developed for large cohorts, sharing approaches that maintain some personalisation despite student number increases.

Limitations

Activity Theory was used for data analysis; an alternative theoretical framework may have yielded different interpretations. The study examines only academics' perspectives rather than including students' viewpoints, providing only one side of the educational exchange. This single perspective approach, whilst allowing for in-depth exploration of teaching practices and perceptions, necessarily limits our understanding of how students experience digital capabilities development.

This study was intentionally contextualised to engineering in one institution with a small number of participants for the purpose of gathering rich data. If the sample had included more participants or those from a wider range of schools within the faculty, the results may have been different. Moreover, the dataset was confined to interviews; had the scope encompassed supplementary data modalities such as video recordings of practice or focus group discussions, further dimensions of the activity system may have been revealed.

The study's focus on education-focused academics as participants means findings illuminate the experiences of teaching-intensive staff. Whilst the systemic contradictions identified are likely relevant across the academic workforce, the specific manifestations and intensity of these tensions may differ for academics in different role types or with different workload profiles. Future research could examine whether the guidance-seeking behaviour and other findings hold across diverse academic roles.

Findings are situated within a specific institutional context and discipline, which may limit transferability to other settings with different structures, resources, or disciplinary practices. The focus on a research-intensive UK university means that findings may not apply directly to teaching-focused institutions or international contexts with different approaches to HE management and delivery.

The emergence of GenAI and its implications for digital capabilities development represents a notable limitation of this study. Although interviews conducted in early 2024 did not capture broad perspectives on GenAI – possibly reflecting the pre-policy institutional context at that time – these technologies are rapidly transforming educational landscapes. This temporal limitation highlights the challenge of researching digital capabilities in a rapidly evolving technological environment.

These limitations are balanced against the study's strengths in providing detailed, contextualised insights into the complex activity systems surrounding digital capabilities development in HE. The alignment between the research questions, theoretical framework, and methodological approach enhances the study's coherence despite these acknowledged limitations.

Conclusion

Summary of key findings and contributions

This paper employs Activity Theory to present a novel analysis of academics' activities in delivering curricula that develop students' practical skills and theoretical knowledge alongside real-world employability, including digital capabilities. Given the limited attention this topic has received, it is crucial to explore how digital capabilities are integrated into the HE engineering curricula, especially against the backdrop of the intensification of academic work.

The findings highlight the diversity of contradictions within this activity system. A central tension emerged between embedding digital capabilities into curricula to the detriment of disciplinary content or requesting that this co-curricular content is delivered by personal academic tutors. What makes this tension significant in engineering education is the technical specificity of required digital capabilities – such as CAD proficiency, computational methods, and simulation techniques –

which are fundamental to professional practice rather than merely supplementary skills. The extensive requirements from engineering accreditation bodies like the Engineering Council and International Engineering Alliance create additional pressures specific to engineering programmes.

This study contributes understanding that workload intensification and the massification of HE have left academics with little time to engage with redesigning teaching activities. Whilst this challenge exists across HE, it manifests distinctively in engineering education due to characteristically large cohort sizes and the hands-on nature of digital capability development in technical disciplines. By centring the investigation on academics whose roles are predominantly oriented towards teaching, the study illuminates how these structural contradictions manifest with particular intensity among staff who carry the heaviest instructional burdens, even as the underlying tensions identified are likely to resonate across the broader academic workforce regardless of role configuration.

Although institutional investment has been made in digital capability tools, awareness and utilisation remain limited. This is problematic in engineering where digital capabilities are not simply generic skills but discipline-specific competencies that require tailored development approaches. Raising awareness of supporting resources may help academics who are struggling to deliver a personalised learning experience to increasing numbers of students.

A further contribution is that both rubrics and module-level learning outcomes need to be more explicit and detailed to produce employable accredited engineering graduates who have a broad range of capabilities. Unlike in less technical disciplines, the implicit expectations about specialised engineering software proficiency create significant barriers to student success and programme effectiveness. The study reveals that these implicit expectations about digital capabilities create barriers to student success and programme effectiveness in ways that are impactful in engineering education, where digital competencies are intrinsically linked to professional practice.

Future research directions

Future research could test the transferability assumptions of this study by applying the Activity Theory framework to examine digital capabilities development in non-STEM disciplines, smaller teaching-focused institutions, or international contexts with different approaches to HE management and delivery. Investigating whether the systemic contradictions and guidance-seeking behaviours identified here manifest similarly across different academic role types would provide valuable insights into whether these are general phenomena or specific to teaching-intensive positions.

Exploring neighbouring activity systems, such as those of students, management, or industry, would also prove valuable, particularly research incorporating student perspectives alongside academic viewpoints to provide a more comprehensive understanding of the digital capabilities development process.

Further studies should examine how GenAI tools are reshaping the contradictions identified in this study, including assessment integrity, pedagogical adaptations, and the evolving definition of digital capabilities in engineering education. Such investigations would provide valuable insights for educational practice during this period of technological transition.

This paper has implications for policy as it identifies that academics want filtering of information and firm guidance from managers to ensure their curricula are appropriate. Such findings contribute to broader discussions about academic autonomy, institutional support, and evolving expectations placed on HE teaching staff in an era of digitalisation and massification. The particular vulnerability of teaching-intensive staff to workload pressures whilst implementing curriculum innovations suggests the need for policy attention to workload allocation, professional development opportunities, and institutional support mechanisms specifically for academics carrying substantial teaching loads.

Author contributions

CRediT: **Tamsyn M. Smith:** Writing – original draft.

Disclosure statement

No potential conflict of interest was reported by the author.

Notes on contributor

Tamsyn M. Smith is Senior Learning Designer Team Lead at the University of Southampton, where she manages a team supporting technology-enhanced learning, curriculum design, and digital capability development across disciplines. With a background in adult basic skills, English, and Media Studies teaching, she has worked in Higher Education since 2012. A strong advocate for Universal Design for Learning (UDL), she has presented nationally on inclusive pedagogy and is progressing through the Learning Designed UDL credentialing pathway. Her achievements include the AdvanceHE CATE Award (2021) and Blackboard Catalyst Award (2020). Tamsyn is a Senior Certified Member of the Association for Learning Technology (SCMALT), an Associate Fellow of AdvanceHE, and is currently undertaking a PhD in E-Research and Technology Enhanced Learning.

ORCID

Tamsyn M. Smith http://orcid.org/0009-0008-9482-601X

References

AAQ (Swiss Agency of Accreditation and Quality Assurance). 2025. *Programme Accreditation*. Aaq.ch. https://aaq.ch/en/accreditation/programme-accreditation/.

Aldhaen, E. 2023. "The Influence of Digital Competence of Academicians on Students' Engagement at University Level: Moderating Effect of the Pandemic Outbreak." *Competitiveness Review* 34 (1), https://doi.org/10.1108/cr-01-2023-0008.

Armah, J. K., and D. V. D. Westhuizen. 2020. "Embedding Digital Capability into the Higher Education Curriculum: The Case of Ghana." *Universal Journal of Educational Research* 8 (2): 346–354. https://doi.org/10.13189/ujer.2020.080203.

Atenas, J., L. Havemann, and C. Timmermann. 2020. "Critical Literacies for a Datafied Society: Academic Development and Curriculum Design in Higher Education." *Research in Learning Technology* 28. https://doi.org/10.25304/rlt.v28.2468.

Austen, L., H. Parkin, S. Jones-Devitt, K. Mcdonald, and B. Irwin. 2016. *Digital Capability and Teaching Excellence: An Integrative Review Exploring What Infrastructure and Strategies Are Necessary to Support Effective Use of Technology Enabled Learning (TEL)*. www.qaa.ac.uk. https://shura.shu.ac.uk/13750/.

Bennett, D., S. Bawa, S. Ananthram, and T. Pitman. 2022. "Is There a Gender Difference in STEM Students' Perceived Employability?" *Education + Training* 64 (6): 754–773. https://doi.org/10.1108/et-01-2021-0029.

Blackmore, J. 2014. ""Wasting Talent"? Gender and the Problematics of Academic Disenchantment and Disengagement with Leadership." *Higher Education Research & Development* 33 (1): 86–99. https://doi.org/10.1080/07294360.2013.864616.

Blau, I., T. Shamir-Inbal, and O. Avdiel. 2020. "How Does the Pedagogical Design of a Technology-Enhanced Collaborative Academic Course Promote Digital Literacies, Self-regulation, and Perceived Learning of Students?" *The Internet and Higher Education* 45:100722. https://doi.org/10.1016/j.iheduc.2019.100722.

Bligh, B., and M. Flood. 2017. "Activity Theory in Empirical Higher Education Research: Choices, Uses and Values." *Tertiary Education and Management* 23 (2): 125–152. https://doi.org/10.1080/13583883.2017.1284258.

Bryson, C. 2004. "What about the Workers? The Expansion of Higher Education and the Transformation of Academic Work." *Industrial Relations Journal* 35 (1): 38–57. https://doi.org/10.1111/j.1468-2338.2004.00299.x.

Bull, S., A. Cooper, A. Laidlaw, L. Milne, and S. Parr. 2024. ""You Certainly Don't get Promoted for Just Teaching": Experiences of Education-Focused Academics in Research-Intensive Universities." *Studies in Higher Education* 50 (2): 1–17. https://doi.org/10.1080/03075079.2024.2333946.

Cheng, M., O. Adekola, J. Albia, and S. Cai. 2021. "Employability in Higher Education: A Review of Key Stakeholders' Perspectives." *Higher Education Evaluation and Development* 16 (1), https://doi.org/10.1108/heed-03-2021-0025.

Cliff, A., S. Walji, R. Jancic Mogliacci, N. Morris, and M. Ivancheva. 2020. "Unbundling and Higher Education Curriculum: A Cultural-Historical Activity Theory View of Process." *Teaching in Higher Education* 27 (2): 1–16. https://doi.org/10.1080/13562517.2019.1711050.

Complete University Guide. 2023. *General Engineering Subject League Table 2024*. www.thecompleteuniversityguide.co.uk. https://www.thecompleteuniversityguide.co.uk/league-tables/rankings/general-engineering.

Davis, F. D. 1989. "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology." *MIS Quarterly* 13 (3): 319–340. https://doi.org/10.2307/249008.

Dixon, E. 2024. "Temporary Employment Rates Twice as High in Academia, Study Finds." *Times Higher Education (THE)*. https://www.timeshighereducation.com/news/temporary-employment-rates-twice-high-academia-study-finds.
Dowling-Hetherington, L. 2014. "The Changing Demands of Academic Life in Ireland." *International Journal of Educational Management* 28 (2): 141–151. https://doi.org/10.1108/ijem-02-2013-0021.
Engeström, Y. 2014. *Learning by Expanding: An Activity-Theoretical Approach to Developmental Research*. Cambridge University Press.
Engineering Council. 2020. *The Accreditation of Higher Education Programmes (AHEP). Fourth edition*. In Engc.org.uk. Engineering Council. https://www.engc.org.uk/our-role-as-regulator/setting-standards/recognising-learning-and-development-programmes.
European Network for Accreditation of Engineering Education (ENAEE). 2024. *EUR-ACE® Framework Standards and Guidelines – ENAEE*. ENAEE. https://www.enaee.eu/eur-ace-system/standards-and-guidelines/.
Flyvbjerg, B. 2006. "Five Misunderstandings about Case-Study Research." *Qualitative Inquiry* 12 (2): 219–245. https://doi.org/10.1177/1077800405284363.
Goswami, D., and J.-B. R. G. Souppez. 2024. "Generative AI in Engineering Education." *2024 UK and Ireland Engineering Education Research Network Annual Symposium, Belfast*. https://research.aston.ac.uk/en/publications/generative-ai-in-engineering-education/.
Gregory, M. S.-J., and J. M. Lodge. 2015. "Academic Workload: The Silent Barrier to the Implementation of Technology-Enhanced Learning Strategies in Higher Education." *Distance Education* 36 (2): 210–230. https://doi.org/10.1080/01587919.2015.1055056.
Griffin, G. 2022. "The "Work-Work Balance" in Higher Education: Between Over-work, Falling Short and the Pleasures of Multiplicity." *Studies in Higher Education* 47 (11): 1–14. https://doi.org/10.1080/03075079.2021.2020750.
Grönlund, A., J. Samuelsson, and J. Samuelsson. 2021. "When Documentation Becomes Feedback: Tensions in Feedback Activity in Learning Management Systems." *Education Inquiry* 14 (2): 194–212. https://doi.org/10.1080/20004508.2021.1980973.
Guest, G., A. Bunce, and L. Johnson. 2006. "How Many Interviews Are Enough? An Experiment with Data Saturation and Variability." *Field Methods* 18 (1): 59–82. https://doi.org/10.1177/1525822X05279903.
Hammond, L. 2019. "The Activity of Student Research: Using Activity Theory to Conceptualise Student Research for Master's Programmes." *Studies in Higher Education* 46 (6): 1055–1067. https://doi.org/10.1080/03075079.2019.1666261.
Hampton, C., D. Reeping, and D. S. Ozkan. 2021. "Positionality Statements in Engineering Education Research: A Look at the Hand That Guides the Methodological Tools." *Studies in Engineering Education* 1 (2): 126. https://doi.org/10.21061/see.13.
Hartman, Y., and S. Darab. 2012. "A Call for Slow Scholarship: A Case Study on the Intensification of Academic Life and Its Implications for Pedagogy." *Review of Education, Pedagogy, and Cultural Studies* 34 (1-2): 49–60. https://doi.org/10.1080/10714413.2012.643740.
Hennink, M. M., B. N. Kaiser, and V. C. Marconi. 2017. "Code Saturation versus Meaning Saturation: How Many Interviews Are Enough?" *Qualitative Health Research* 27 (4): 591–608. https://doi.org/10.1177/1049732316665344.
Houston, D., L. H. Meyer, and S. Paewai. 2006. "Academic Staff Workloads and Job Satisfaction: Expectations and Values in Academe." *Journal of Higher Education Policy and Management* 28 (1): 17–30. https://doi.org/10.1080/13600800500283734.
International Engineering Alliance. 2021. *Graduate Attributes & Professional Competencies*. https://www.ieagreements.org/assets/Uploads/IEA-Graduate-Attributes-and-Professional-Competencies-2021.1-Sept-2021.pdf.
International Engineering Alliance. 2025. *Washington Accord*. International Engineering Alliance. https://www.internationalengineeringalliance.org/accords/washington-accord.
Jisc. 2019. *What is Digital Capability?* JISC. https://digitalcapability.jisc.ac.uk/what-is-digital-capability/.
Jones, B., A. Pilot, Pierre van Eijl, and J. Lappia. 2020. "The w-Shaped Model of Professional Competencies for the Fourth Industrial Revolution and Its Relevance to Honors Programs." *Journal of the European Honors Council* 4 (1), https://doi.org/10.31378/jehc.123.
Jugembayeva, B., and A. Murzagaliyeva. 2022. "Physics Students' Innovation Readiness for Digital Learning within the University 4.0 Model: Essential Scientific and Pedagogical Elements That Cause the Educational Format to Evolve in the Context of Advanced Technology Trends." *Sustainability* 15 (1): 233. https://doi.org/10.3390/su15010233.
Kenny, J. 2016. "Academic Work and Performativity." *Higher Education* 74 (5): 897–913. https://doi.org/10.1007/s10734-016-0084-y.
Kinman, G., and F. Jones. 2003. ""Running up the down Escalator": Stressors and Strains in UK Academics." *Quality in Higher Education* 9 (1): 21–38. https://doi.org/10.1080/13538320308162.
Klochkova, E., Y. Serkina, V. Prasolov, and V. Movchun. 2020. "The Digitalisation of the Economy and Higher Education." *Space and Culture India* 7 (4): 70–82. https://doi.org/10.20896/saci.v7i4.697.
Larsen, E., and R. Brandenburg. 2023. "Navigating the Neo-Academy: Experiences of Liminality and Identity Construction Among Early Career Researchers at One Australian Regional University." *The Australian Educational Researcher* 50:1069–1087. https://doi.org/10.1007/s13384-022-00544-1.

Lewis, J. 2023. "Students and the Rising Cost of Living." *House of Commons Library*. https://commonslibrary.parliament.uk/research-briefings/cbp-9886/.

Lincoln, Y. S., and E. G. Guba. 2016. *The Constructivist Credo*. Routledge. https://ebookcentral.proquest.com/lib/lancaster/reader.action?docID=1187038#.

Malterud, K., V. D. Siersma, and A. D. Guassora. 2015. "Sample Size in Qualitative Interview Studies: Guided by Information Power." *Qualitative Health Research* 26 (13): 1753–1760. https://doi.org/10.1177/1049732315617444.

Masenya, T. M. 2021. "Digital Literacy Skills as Prerequisite for Teaching and Learning in Higher Education Institutions." *Mousaion: South African Journal of Information Studies* 39 (2), https://doi.org/10.25159/2663-659x/8428.

McCance, K. R., S. D. Teeter, M. R. Blanchard, and R. A. Venditti. 2023. "Using Activity Theory to Understand the Interactions of a University Interdisciplinary Team of Scientists and Science Educators." *Studies in Higher Education* 48 (6): 892–909. https://doi.org/10.1080/03075079.2023.2172564.

Miles, M. B., A. M. Huberman, and J. Saldaña. 2020. *Qualitative Data Analysis: A Methods Sourcebook*. Sage.

Mishra, P., and M. J. Koehler. 2006. "Technological Pedagogical Content Knowledge: A Framework for Teacher Knowledge." *Teachers College Record* 108 (6): 1017–1054. https://doi.org/10.1111/j.1467-9620.2006.00684.x.

Molnárfi, L. 2022. *Trinity: Stop Putting Our Teachers on Precarious Contracts! – Students4Change*. Students4Change. https://students4change.eu/2022/09/03/trinity-stop-putting-our-teachers-on-precarious-contracts/.

Morgan, A., R. Sibson, and D. Jackson. 2022. "Digital Demand and Digital Deficit: Conceptualising Digital Literacy and Gauging Proficiency among Higher Education Students." *Journal of Higher Education Policy and Management* 44 (3): 1–18. https://doi.org/10.1080/1360080x.2022.2030275.

Murray, M., J. Pérez, and J. Fluker. 2022. "Digital Literacy in the Core: The Emerging Higher Education Landscape." *Issues in Informing Science & Information Technology* 19:1–13. https://doi.org/10.28945/4957.

Mwanza-Simwami, D. 2011. "AODM as a Framework and Model for Characterising Learner Experiences with Technology." *Journal of E-Learning and Knowledge Society* 7 (3): 75–85. https://doi.org/10.20368/1971-8829/553.

New Jersey Institute of Technology (NJIT). 2025. *ASIIN*. Accreditation.org. https://accreditation.org/accreditation-processes/accrediting-bodies/asiin.

Organisation for Economic Co-operation and Development. 2024. *The State of Academic Careers in OECD Countries – An Evidence Review* (OECD Education Working Paper No. 91). https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/02/the-state-of-academic-careers-in-oecd-countries_dc11b29d/ea9d3108-en.pdf.

Ostanina, A., O. Bazyl, O. Tsviakh, and N. Dovzhuk. 2023. "Formation of Digital Competence in Higher Education Students as a Basis for the Transformation of Education of the Future." *Futurity Education* 3 (1): 126–135. https://doi.org/10.57125/fed.2023.25.03.10.

Otoo, B. K. 2020. "Declaring my Ontological and Epistemological Stance: A Reflective Paper." *The Journal of Educational Thought (JET) / Revue de La Pensée Éducative* 53 (1): 67–88. https://doi.org/10.55016/ojs/jet.v53i1.71097

Papadopoulos, A. 2017. "The Mismeasure of Academic Labour." *Higher Education Research & Development* 36 (3): 511–525. https://doi.org/10.1080/07294360.2017.1289156.

Park, Y., and I.-H. Jo. 2016. "Using Log Variables in a Learning Management System to Evaluate Learning Activity Using the Lens of Activity Theory." *Assessment & Evaluation in Higher Education* 42 (4): 531–547. https://doi.org/10.1080/02602938.2016.1158236.

Qadir, J. 2023. "Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education." *2023 IEEE Global Engineering Education Conference (EDUCON)*, 1–9. https://doi.org/10.1109/educon54358.2023.10125121.

Quality Assurance Agency for Higher Education. 2023. *Subject Benchmark Statement – Engineering* (5th edition). Www.qaa.ac.uk; Quality Assurance Agency for Higher Education. https://www.qaa.ac.uk/the-quality-code/subject-benchmark-statements/subject-benchmark-statement-engineering.

Ramírez-Montoya, M. S., M. I. Loaiza-Aguirre, A. Zúñiga-Ojeda, and M. Portuguez-Castro. 2021. "Characterization of the Teaching Profile within the Framework of Education 4.0." *Future Internet* 13 (4): 91. https://doi.org/10.3390/fi13040091.

Saffron Powell, S., and T. Varga-Atkins. 2013. *Digital Literacies: A Study of Perspectives and Practices of Academic Staff*. SEDA. https://www.seda.ac.uk/resources/files/PowellVargaAtkinsFinalRpt.pdf.

Saienko, V., N. Kurysh, and I. Siliutina. 2022. "Digital Competence of Higher Education Applicants: New Opportunities and Challenges for Future Education." *Futurity Education* 2 (1): 37–46. https://doi.org/10.57125/fed/2022.10.11.23.

Scott, C. L. 2015a. *The Futures of Learning 2: What Kind of Learning for the 21st Century?* Unesco.org. https://unesdoc.unesco.org/ark:/48223/pf0000242996.

Scott, C. L. 2015b. *The Futures of Learning 3: What Kind of Pedagogies for the 21st Century?* Unesco.org. https://unesdoc.unesco.org/ark:/48223/pf0000243126.

Sissons, H., and D. Mulrennan. 2022. "Technology-Enhanced Learning in Designing for Uncertainty." *Pacific Journal of Technology Enhanced Learning* 4 (1): 42–43. https://doi.org/10.24135/pjtel.v4i1.137.

Stake, R. E. 1995. *The Art of Case Study Research*. Thousand Oaks, CA: Sage Publications.

Tessema, K. A. 2009. "The Unfolding Trends and Consequences of Expanding Higher Education in Ethiopia: Massive Universities, Massive Challenges." *Higher Education Quarterly* 63 (1): 29–45. https://doi.org/10.1111/j.1468-2273.2008.00408.x.

Tham, T. L., and P. Holland. 2018. "What Do Business School Academics Want? Reflections from the National Survey on Workplace Climate and Well-Being: Australia and New Zealand." *Journal of Management & Organization* 24 (4): 492–499. https://doi.org/10.1017/jmo.2018.3.
The Guardian. 2023. "The Guardian University Guide 2024 – The Rankings." *The Guardian*. https://www.theguardian.com/education/ng-interactive/2023/sep/09/the-guardian-university-guide-2024-the-rankings.
Times Higher Education. 2023. *Best Universities for Engineering Degrees in the UK 2023*. THE Student. https://www.timeshighereducation.com/student/best-universities/best-universities-engineering-degrees-uk.
Traxler, J. 2018. "Digital Literacy: A Palestinian Refugee Perspective." *Research in Learning Technology* 26. https://doi.org/10.25304/rlt.v26.1983.
University and College Union (UCU). 2016. *Precarious Work in Higher Education: A Snapshot of Insecure Contracts and Institutional Attitudes*. https://www.ucu.org.uk/media/7995/Precarious-work-in-higher-education-a-snapshot-of-insecure-contracts-and-institutional-attitudes-Apr-16/pdf/ucu_precariouscontract_hereport_apr16.pdf.
University of Southampton. 2024. *Our Goals*. University of Southampton. https://www.southampton.ac.uk/about/strategy/goals.
Van Essen-Fishman, L., D. Morgan, and A. Madrigal. 2023. *Getting Real About Graduate Earnings*. www.hesa.ac.uk. https://www.hesa.ac.uk/insight/28-11-2023/getting-real-about-graduate-earnings.
Varga-Atkins, T. 2020. "Beyond Description: In Search of Disciplinary Digital Capabilities through Signature Pedagogies." *Research in Learning Technology* 28. https://doi.org/10.25304/rlt.v28.2467.
Vostal, F. 2015. "Speed Kills, Speed Thrills: Constraining and Enabling Accelerations in Academic Work-Life." *Globalisation, Societies and Education* 13 (3): 295–314. https://doi.org/10.1080/14767724.2014.959895.
Wells, M. R. 2023. "Embedding Digital Fluency in Our Courses: Moving from Theory to Practice." *Collected Essays on Learning and Teaching* 14 (1), https://doi.org/10.22329/celt.v14i1.7142.
Wenger, E. 1998. *Communities of Practice: Learning, Meaning, and Identity*. Cambridge: Cambridge University Press.
Wolf, A., and A. Jenkins. 2021. *Managers and Academics in a Centralising Sector*. Nuffield Foundation. https://www.nuffieldfoundation.org/wp-content/uploads/2021/12/Managers-and-academics-in-a-centralising-sector.pdf.
World Economic Forum. 2024. *Reskilling Revolution: Education 4.0*. Initiatives.weforum.org. https://initiatives.weforum.org/reskilling-revolution/education-4-0.

Appendices

Appendix A. Semi-structured interview guide

What is your role at the university? How long have you been doing it? Do you feel the role has changed since you first started in terms of the workload?

(1) Tools:
- What resources or technologies do you use to enhance student digital capabilities in your teaching?
- Can you explain how these tools are utilised in achieving your teaching objectives?
- Are there any specific platforms or software you rely on for assessing or enhancing student digital skills?
- Further topics:
  - Specific digital platforms or technologies used in teaching.
  - Time-saving tools or techniques employed amidst increased workloads.

(2) Rules:
- Are there any institutional policies or guidance, or disciplinary or professional norms that influence your approach to developing student digital capabilities?
- What institutional policies or guidelines influence your approach to integrating student digital capabilities into your teaching?
- How do academic standards and curriculum requirements impact your decision-making process in teaching design?
- Are there any disciplinary or professional norms that guide your use of digital tools in teaching?
- Further topics:
  - Compliance with curriculum standards amidst workload intensification.
  - Adaptation of teaching methods due to policy changes.

(3) Division of Labour:
- Could you describe the collaboration or division of responsibilities among faculty members when it comes to addressing student digital capabilities?
- How do you collaborate with learning designers, educational developers, or other colleagues to support student digital skill development?

- Are there any departmental structures or support mechanisms in place to facilitate the integration of digital capabilities into teaching?
- Further topics:
  - Collaborative efforts to manage workload intensification.
  - Delegation of tasks related to addressing student digital capabilities.

(4) Community and Tools:
- How does the academic community collectively support the development of student digital capabilities?
- Are there any community-led initiatives or resources available to faculty for enhancing digital skills among students?
- Do you engage in communities of practice or professional networks that focus on leveraging digital tools for teaching improvement?
- Further topics:
  - Shared tools or platforms for collaboration amidst workload intensification.
  - Accessibility of resources to support teaching objectives in the context of increased workload.

(5) Community and Rules:
- What institutional or community-wide policies shape the approach to integrating student digital capabilities across teaching practices?
- How do departmental or institutional cultures influence the emphasis on digital skill development in teaching?
- Are there any shared beliefs or values within the academic community regarding the importance of digital literacy for students?
- Further topics:
  - Impact of institutional policies on workload distribution within the academic community.
  - Collaboration and rule adherence amidst workload intensification.

(6) Community and Division of Labour:
- How does collaboration among faculty members contribute to the collective effort of addressing student digital capabilities?
- Are there any formal or informal roles within the academic community that focus specifically on enhancing digital skills among students?
- How do departmental structures or leadership initiatives support the coordination of efforts in addressing student digital capabilities?
- Further topics:
  - Strategies for workload sharing and collaboration within the academic community.
  - Adaptation of *Division of Labour* practices to address workload intensification.

Appendix B. Structured Activity Theory codebook

This structured codebook was developed based on Activity Theory's key components (*Subject*, *Tools*, *Object*, *Rules*, *Community*, *Division of Labour*), and Engeström's (2014) four levels of contradictions. Each code includes a definition, example indicators from the data, application guidelines to ensure consistent analysis, and participant quotes that illustrate the code in practice. This approach provides a rigorous framework for exploring the complex interrelationships within academic practice related to digital capabilities development in higher education.

Coding framework

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
Subject	The individual(s) or group(s) carrying out the activity being studied	• References to academics, their roles, identities, backgrounds, motivations	Use when participants discuss their roles, experiences, or perspectives as teachers. Do not use when discussing students or administrators who are not directly engaged in teaching.	'I'm a senior teaching fellow in the School of Engineering. I spend most of my time looking after the engineering foundation year'. 'So, my role is Design Service Experimental Officer, which involves offering design advice for everybody in the School of Engineering – particularly design-for-manufacture so that people can understand how to specify designs suitably so they can be made and manufactured'. 'At the moment I am a lecturer in in computer science, and I have been in this role since May 2020'. 'I'm an associate professor in Energy and Building on a balanced pathway. So, I've got three jobs: I teach … Then I research … And then as well, I've got a leadership role'.
Tools	Physical or conceptual instruments used to mediate activities	Digital learning environments: • Virtual Learning Environments (Blackboard, ECS Notes) • Microsoft Teams for communication and collaboration • Wikis • EdShare repository • Software for disciplinary skills: • CAD software (SolidWorks, AutoCAD) • Programming tools (Python, IDLE, R) • Materials analysis (Granta EduPack) • Simulation software Assessment and feedback tools: • Rubrics and marking criteria • Digital logbooks (OneNote) • Online assessment platforms • Blackboard Collaborate for group presentations Self-assessment and skills development resources: • Jisc Discovery Tool • LinkedIn Learning	Use when tools for teaching or developing digital capabilities are mentioned. Do not use when discussing resources not directly used in teaching activities	'We rely heavily on Blackboard, obviously, to store information and communicate with our very large cohorts … We've also integrated the use of Teams quite heavily to be able to communicate more focused … We do use Granta EduPac, which is to help them understand better the use of materials, material properties, the impact of material choices … There's also SolidWorks, which is 3D CAD'. (Della) 'In the coursework module, a computer programming element, the first four weeks are designed to introduce them to concepts of digital literacy. So that's where they use the Discovery Tool. They also learn about LaTeX … But they also learn about some general information like LinkedIn, cloud storage, so, they're introduced to OneDrive and Google Drive and sorting out those elements there'. (Chris) 'I do use R as a software. And they have to complete basically an edit analysis in class … I use quite a lot … I know Blackboard Collaborate is being discontinued, but actually use quite a bit, Blackboard Collaborate, especially for group work'. (Felicity) 'These days we kind of expect the ability to use the software as a means to achieving a broader end within engineering. Similarly, we ask people to make presentations, and we often enjoy their output because they've been able to illustrate it beautifully,

Continued.

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
		• Digital badges (ThingLink) Teaching methodologies and materials: • Real-life scenarios and case studies • Workshop materials and templates • Video tutorials and recorded lectures • Group project frameworks		set it out in a PowerPoint or equivalent slide format, and illustrate it with video … ' (Alex)
Object	The purpose or objective that drives the activity	• References to graduate employability, digital skills development, learning outcomes Object: (idealised aim)Short term: Delivering a tripartite curriculum • practical/technical • theoretical • real-world employability Long-term:producing employable accredited engineering graduates who have a broad range of capabilities*Outcome:* (actual results – can be unintended) • Students are strategic about their learning and focused on what they need to do to pass	Use when discussing goals or intended outcomes of teaching activities. Do not use when discussing unintended consequences.	'As long as I've been involved in higher education, it's always been about, 'OK, how useful is this person actually going to be in the real world?' And that's always been a real focus of what I've done'. (Eric) 'My philosophy, I suppose, is very much trying to get everyone on the same technical working level … I very much want them to – when they pass foundation year – to go into their Part One with the skills that they need to be able to do the rest of their degree. But we do go a bit beyond that. We do think about long-term employability'. (Chris) 'MSc, definitely employability is key. They are there for one year. It's a very short time at university and they will have had kind of like the base skills and knowledge at undergraduate level. This one is more like a specialist … We are in engineering as well, so it's very much employability focused'. (Felicity) 'In our department, which has recently undergone a curriculum review, the philosophy or the main aim, or at least what we advertise, is that our curriculum and our offerings address three main aspects. One is the technical and practical aspect … a very theoretical aspect … And a third aspect, which is the employability and real-world aspect'. (Boris)
Rules	Formal and informal regulations, norms, and conventions	Professional accreditation requirements: • Engineering Council's AHEP4 framework • British Computer Society standards • EEE (Institute of Engineering and Electrical Engineering) guidelines • JBM (Joint Board of Moderators) accreditation criteria Institutional policies and expectations:	Use when institutional or professional guidelines are mentioned. Do not use when discussing personal preferences not linked to external requirements.	'We have the AHEP4 guidelines in engineering. The AHEP4 matrices definitely have an impact on what we deliver. There's a big conversation at the moment, and has been for years really, but there seems to be more of a push here than there was in previous institutions on sustainability, and on equality, diversity, and inclusion'. (Della) 'In AHEP4 there is quite a bit about the student's ability to provide feedback to review others … I'm trying to think of what's specifically about digital capabilities,

Continued.

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
		• University teaching quality standards • Faculty-level educational strategies • School/departmental teaching expectations • Module validation and review processes Industry-influenced standards: • Industrial Advisory Board recommendations • Employment sector expectations • Digital capabilities required for workplace readinessAssessment regulations and practices: • Academic integrity requirements • Marking criteria and moderation processes • Student attendance policies • Digital submission requirementsImplicit pedagogical norms: • Expectations that skills development is embedded in the curriculum • Consensus on appropriate teaching methodologies • Acceptable levels of digital integration in learning activities		and I'm sure that is … I'll have to triple check it, but there's stuff on there, technical literature, and how they can go and find those resources and evaluate it'. (Chris) 'So yes, there are a set of bodies that influence the way we design our modules because they have to go through approval processes, and one of them is British Computer Society, another one is the IEEE (Institute of Engineering and Electrical Engineering), so there are a few external bodies that accredit what we teach'. (Boris) 'So, the Engineering Council has an accreditation of higher engineering programmes document … So, anything that the Engineering Council, representing the interests of professional engineers, wish the graduates seeking employment in engineering to be capable of goes into the AHEP (the accreditation of higher education programmes) document which gets updated from time-to-time'. (Alex)
Community	The social context within which the activity takes place	Internal academic community: • Teaching teams and colleagues • Departmental staff • School/faculty academic networks • Education committeesStudent community: • Diverse cohorts with varying digital skills • Students with/without prior experience • Mature students with industry experience • International studentsProfessional services: • Library staff • Academic Skills Hub • Digital Learning teams • Technical support staffExternal professional community:	Use when the social environment affecting teaching is discussed. Do not use when focusing solely on individual actions without social context.	'We are with the embedded Skills team, which is part of the Academic Skills Hub. So, it's mostly Alison Daniel who has been helping us embed some academic skills into Routes to Success, which is by itself an academic skills module'. (Chris) 'We have very good industrial Advisory Board, and they are … we're quite self-reflective on what needs to happen to deliver what the outcome is'. (Felicity) 'We are a diverse group of people who have all sorts of different interests working in different places'. (Alex) 'We have access to various networks and teams across the university, so Digital Learning Connect, CHEP Community, others like the Digital Learning team (which I suppose is Digital Learning Connect). Yeah, various teams that we probably would go to if we had a question'. (Chris)

Continued.

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
		• Professional bodies (IMechE, IEEE, BCS) • Industrial Advisory Boards • AdvanceHE networks • Employer partners		'It's a multidisciplinary group of students as well, especially at master level'. (Felicity)
Division of Labour	How tasks are distributed among community members	Academic staff responsibilities: • Teaching and assessment duties • Course design and updates • Professional development requirements • Administrative and reporting tasks • Research, alongside teaching commitments,Management and leadership roles: • Programme leaders setting curriculum direction • Education committee decision-making • Deputy Heads are filtering information to teaching staff • Quality assurance monitoring and feedbackStudent responsibilities: • Attendance at scheduled sessions • Independent skills practice • Collaboration using digital platforms • Engagement with learning resourcesProfessional services contributions: • Library providing research skills training • Technical staff supporting specialist software • Delivering student support • Administrative staff processing assessments	Use when task distribution or responsibilities are mentioned. Do not use when discussing individual skills or capabilities.	'I'm pretty sure the reason students come to Southampton is that it's a Russell Group university. I think we are not really that inventive in terms of our delivery, and we're certainly very much in the twentieth century with our assessment. So, I don't think any of those elements would encourage students to come and study with us'. (Della) 'Many of our staff are directed towards their various responsibilities and haven't been directed towards being responsive to these types of questions. And it might also be, if there were a pool of staff that have one hour a week, who do people with questions go to?' (Alex) 'I don't know why I got a design degree because I actually don't really need it. What I actually need is an admin degree because I spend all my time filling in spreadsheets and doing Word documents and all that sort of stuff'. (Eric) 'I think it fluctuates very much depending on what particular area you're looking at. As I say, I've been amazed by the lack of management that there is. The lack of supervision there is over certain things'. (Eric) 'I work in pair with [colleague 1] and [colleague 2] and we all three quite approachable maybe, so student just tend to come and chat to us, you know'. (Felicity)
Primary Contradictions	Tensions within a single component of the activity system	• References to conflicts within tools (e.g. software that is both helpful and limiting)	Use when tensions exist within one node of the activity triangle. Do not use when tensions exist between different components.	'As soon as you ask a student to download a programme … It's like, 'Oh, I don't have the right version of Windows', or 'I don't have that as I have a Mac,' and blah blah blah, and it's just an absolute nightmare, and you spend your time dealing with the hardware rather than with the learning, and it's just a distraction … ' (Felicity) 'The software enables you to do that fairly easily once

Continued.

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
				you know how to do it, but you do need practice – there's a lot of quite a lot of practice needed – and a bit of an instinct for working out what you can do in the software. If you don't know something, can you work it out within the software?' (Alex) 'Our VLE is, yeah, we have different VLEs to use. So, as you know, in ECS, we have our own system: ECS Notes. Yeah, we also use wiki a lot, yeah, wiki pages, and obviously the university default VLE, Blackboard, yeah. And in some modules, it's used; in some others, it's not used'. (Boris)
Secondary Contradictions	Tensions between components of the activity system	• References to conflicts between Rules and tools, or between *Object* and *Division of Labour*	Use when tensions exist between two different components. Do not use when tensions exist within one component or across activity systems.	'I suppose the biggest challenge is just the scale of it more than anything else. It's how do you effectively teach a cohort of nearly 600 students? … each year I'm tinkering and playing around trying to think of new methods to give what feels like a valuable one-to-one experience to such a huge number of students'. (Eric) 'I think we assume that they do have much better digital capabilities than we have, in a way, because they've grown up swiping. We haven't. But no, I mean they are digitally very capable. I think, especially in the lower years, what we are overestimating is their ability to find information. So, their search capacity is not developed yet'. (Della) 'The design of teaching and learning activity? So, it depends on the focus of each group … The challenges on that was to make sure that you had very clear alignment year-on-year progression between the content, the assessment, and the actual learning that the student needed to use that core skill for semester 2 modules and their dissertation and future life'. (Felicity) 'There would be an element which isn't very rigorously set into a marking rubric where we would hope we can expect reasonable quality, but there isn't a kind of specific guidance for 'these are the things we're going to mark against; please get these specific things right in that in that respect''. (Alex)
Tertiary Contradictions	Tensions between different versions of the same activity system	• References to conflicts between old and new models of teaching	Use when discussing tensions between traditional and emerging approaches. Do not use when discussing	'It's a strange disconnect between I feel our generation and that generation that the way we use the digital media is very different and they can't quite get their heads around VLEs and e-mail to the same degree that we can, which is an odd thing to say because we

Continued.

Code	Definition	Example indicators	When to use/when not to use	Example participant quotes
			tensions within current practice only.	weren't brought up on emails and VLE'. (Eric) 'Email seems to be a method of communication they just aren't happy with. I don't know where that comes from. And I will say to them, 'Well, have you emailed your tutor to try and sort this out?' If they've got a problem, it's like, 'Oh, I don't really want to e-mail anybody. No. Don't want to bother them''. (Eric) 'AI is a huge issue on this type, so ChatGPT, it really changed the assessment structure, so I shifted to … so it used to be a coursework and an in-person exam, to then a coursework and an online exam for COVID, and then a coursework and a Viva to be able to actually assess that that is their work'. (Felicity)
Quaternary Contradictions	Tensions between the activity system and those neighbouring it	• References to conflicts between teaching and administrative systems	Use when discussing tensions between different activity systems. Do not use when discussing tensions within one activity system.	'I don't really have anyone saying to me, 'You need to make sure you do it. They're doing this. But I think we could do that bit'. But I think that needs to be a bit more better in the school. It would help me if some people would say, ' … they need to be able to do XYZ''. (Chris) 'I don't think they're in a good position to. Yeah, they're not specialists in energy and building, so therefore, they can't really substantively comment on the nature of what's … They can, surely, you know, have an opinion on the method of teaching, but not the substantive nature of it, because their specialism is elsewhere'. (Felicity, regarding input from Associate Deans) 'I was quite shocked how and undeveloped those skills are when students come into university considering that they've spent at least a couple of years during their A levels having to do some level of research or some level of information gathering and revision'. (Della) 'You know, I actually want somebody to be looking at that stuff, saying, 'OK, you're doing it right. You're doing it wrong. You should need to change this. You need to do this'. But that seems to be pretty much absent, which is a new experience for me'. (Eric)

Additional important codes

Code	Definition	Example indicators	When to use/When not to use	Participant quotes
Workload intensification	References to increasing demands, student numbers, or administrative tasks	Expressions of being time-pressed, having limited capacity to engage with additional tasks, or increased administrative burden	Use when participants refer to growing pressures on their time or workload. Do not use when discussing routine job expectations.	'I suppose the design team is probably a particularly short-staffed area at the moment, but there's very little time for reflection in any way shape or form, let alone for professional development'. (Della) 'I think my interpretation is I know what I'm doing for most of it, but then sometimes someone will say, 'Oh, this happens elsewhere', and I'll be like, 'Oh my God, that's crazy!' And I'll be like, 'Oh, I never thought of that!'' (Chris) 'I get emailed all the time. I get 100 emails in my inbox. And I answer every single one of them straight away'. (Eric) 'When I first started eight years ago, it had about 30 students on it. It now has 86 students. So how do you deliver at scale the same learning outcome is challenging'. (Felicity)
Guidance seeking	Expressions of desire for clearer direction or management guidance	References to wanting more explicit instructions, policies, or leadership guidance	Use when participants express a need for more direction. Do not use when participants discuss autonomous decision-making processes.	'You know, I actually want somebody to be looking at that stuff, saying, 'OK, you're doing it right. You're doing it wrong. You should need to change this. You need to do this'. But that seems to be pretty much absent, which is a new experience for me'. (Eric) 'But I will say the content will be more on feeding back to I would say industry. There is a lot of self-reflection, less on masters, but more on undergraduate. We have very good industrial Advisory Board, and they are … we're quite self-reflective on what needs to happen to deliver what the outcome is'. (Felicity) 'We haven't traditionally taught, specifically, how to keep things secure. So that is, you know, the security of data, data security. Cyberwars potentially are more of an issue. Wouldn't it be sensible for all engineers to some degree, depending on their role, to be aware of any security issues in the outcome of their engineering … ' (Alex)